Element H

To ensure a proper, in-depth analysis of our final product, the following is our group's testing procedure. We previously talked through the process with Doctor Bill Rodriquez, who gave us good feedback on our design.

1. Preparation Phase:
    - Gather all necessary materials and equipment for the testing procedure.
    - Ensure the detachable lid and barrel system is securely assembled and ready for testing.
    - Set up the testing environment in a safe and controlled area.
2. Baseline Measurements:
    - Measure the initial weight and dimensions of the barrel system.
    - Record the time it takes to securely attach the lid to the barrel.
3. Functional Testing:
    - Place food items or treats inside the barrel to simulate the feeding process.
    - Securely attach the lid to the barrel.
    - Observe and record the time it takes for the bears (or bear-mimicking mechanism) to successfully open the lid and access the food.
    - Repeat this process multiple times to ensure consistency and reliability of the lid and barrel system.
4. Durability Testing:
    - Subject the lid and barrel system to simulated bear-like forces using controlled mechanical pressure or weight.

- Gradually increase the pressure or weight to assess the durability and strength of the system.
- Record any signs of damage, deformation, or failure of the components.

5. Ease of Access Testing:
    - Invite individuals of varying ages and strengths (including the high schoolers working on the project) to attempt opening and closing the lid.
    - Record any difficulties, challenges, or ease of use encountered during the process.
    - Collect feedback on the ergonomics and usability of the detachable lid and barrel system.
6. Data Analysis:
    - Compile all recorded measurements, observations, and feedback.
    - Analyze the effectiveness of the design based on factors such as:
        - Time taken to open/close the lid.
        - Durability under simulated bear-like forces.
        - Ease of access for users.

<u>Evaluation:</u>

The preparation phase ensures that all materials and equipment are gathered and the testing environment is safe and controlled. This step is critical to guarantee consistent and reliable results. The feasibility of this phase is high, as it requires standard procedures that can be readily implemented.

Measuring the initial weight and dimensions of the barrel system and recording the time to attach the lid provides a control against which other tests can be compared. This step is feasible and straightforward, requiring basic measurement tools and timing devices.

Placing food items inside the barrel and observing how long it takes for bears or a bear-mimicking mechanism to access the food simulates real-world use. Repeating this process multiple times ensures the reliability of the system. This test is feasible, although using live bears would require coordination with a zoo and adherence to animal welfare guidelines. Using a bear-mimicking mechanism, such as a mechanical arm or weighted dummy, could be a more practical alternative.

Simulating bear-like forces with controlled mechanical pressure or weight is feasible with the appropriate equipment, such as a hydraulic press or weighted impact testing setup. This test assesses the structural integrity of the design under stress, which is critical for ensuring the system's robustness. Recording signs of damage or deformation will provide objective data on durability.

Inviting individuals of varying ages and strengths to test the lid's usability provides insight into the ergonomics and user-friendliness of the design. This step is highly feasible, as it involves human participants who can offer valuable feedback. Ensuring a diverse group of testers will enhance the validity of the results.

Compiling and analyzing recorded measurements, observations, and feedback allows for a comprehensive evaluation of the design's effectiveness. This step is feasible and essential for interpreting the collected data and making informed decisions about potential improvements.

The testing plan addresses all high-priority design requirements by providing a logical and well-developed explanation of how testing would yield objective data regarding the effectiveness of the design. The tests are feasible within the instructional context, employing both physical and mechanical modeling where appropriate. Each phase of the plan is designed to collect specific, relevant data that can be analyzed to assess the design's performance.

Seeking professional help could have provided additional validation, however, the timeframe did not allow for immediate communication between an expert and the group. This could have been prevented if help was requested prior to that of the making of this element. Another solution could have been consulting a matrices of opinions by consulting a general opinion of the design and device performance.

The testing plan is feasible and well-structured to meet the design requirements. It comprehensively addresses the necessary aspects of preparation, functionality, durability, ease of access, and data analysis. However, this project could have proved greater if proper execution and expert input have been further implemented. That being said, the device testing procedure is reinforced with logical and through reasoning, providing less room for bias and error to be implemented into the collection of the resulting data of the procedure.